

Hybrid MCMC and Physics-Aware Transformer-Based Inversion of Layered Hydrologic Storage from InSAR Deformation: Case Study of Emilia-Romagna, Italy

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Abstract—A novel hybrid machine-learning framework based on MCMC and Swin3D U-Net transformers is presented to directly invert layered hydrologic storage from InSAR time series of land deformation. The motivation is the need for groundwater-related storage estimates at spatial scales relevant to aquifer systems, where existing large-scale water-storage products remain too coarse and direct subsurface observations are sparse. However, the observed InSAR signal is often noisy, spatially heterogeneous, and influenced by non-hydrologic processes. To address this, the proposed framework combines deformation-space Bayesian inversion, forward hydro-mechanical consistency, and transformer-based residual learning within a multisensor setting. The framework is first evaluated in a synthetic five-layer setting to test identifiability under controlled conditions, and is then applied to Emilia-Romagna, Italy, where the strongest supported real-data result is obtained through a grouped balanced multisensor Stage 1 formulation constrained by SMAP, with SWOT included as an exploratory surface-water constraint and groundwater wells used for independent validation, summaries over the shared period 2017-01-04 to 2024-08-01. The strongest supported real-data result is the grouped posterior state $[S_0 + S_s, S_d + S_r, S_g]$, which remains physically plausible in magnitude and is independently supported by groundwater-well anomalies across the Bologna monitoring network. Independent validation against Bologna groundwater wells further supports the grouped groundwater signal, with 83 evaluated station series, a median best-match anomaly correlation of about 0.704, and a conservative trusted subset of 21 stations across 6 hydrogeologic groups. To our knowledge, this is the first approach to directly invert layered water storage from land deformation without requiring prior explicit isolation of a purely groundwater-related signal. These results show that high-resolution layered hydrologic storage can be inferred from mixed InSAR deformation when embedded in a hybrid MCMC and physics-aware transformer-based framework.

Index Terms—InSAR deformation, layered soil water content, groundwater storage, hydrologic storage inversion, Bayesian state-space modeling, Markov chain Monte Carlo, physics-aware transformers, Swin3D U-Net, multisensor assimilation.

I. INTRODUCTION

WATER security increasingly requires monitoring systems that can resolve not only basin-scale mass change, but also compartment-level and aquifer-scale storage dynamics in regions affected by irrigation, drought, and land subsidence. Groundwater change is directly tied to agricultural

productivity, drought resilience, environmental management, and, in stressed aquifer systems, to compaction and subsidence [1]–[3]. Emilia-Romagna is a suitable case study because it is a highly monitored multilayer aquifer system with strong groundwater variability driven by interacting surface-water, shallow-aquifer, and deeper confined-aquifer dynamics, while groundwater is also heavily used for agricultural irrigation and other regional water demands [4]–[6].

This need has intensified in the current Earth-observation era, where gravimetry, hydrologic models, soil-moisture products, surface-water missions, and dense radar deformation measurements coexist, but differ strongly in spatial support and physical meaning.

At continental to regional scales, GRACE and GRACE-FO have transformed observation of terrestrial water storage (TWS), but their effective support remains too coarse for direct aquifer-scale diagnosis in heterogeneous and intensively managed settings [7]–[9]. Bayesian data-assimilation frameworks have therefore been developed to fuse large-scale observations with hydrologic models. In particular, Mehrnegar et al. formulated a Markov Chain Monte Carlo data-assimilation framework (MCMC-DA) that integrates GRACE total water storage changes with W3RA to update groundwater and soil-water storage components [7], using a time-varying estimation compared to earlier DMDA approaches [10]–[15]. This provides a rigorous and uncertainty-aware assimilation model, but it is built for a setting in which observations and model outputs are both storage-related quantities.

The InSAR setting is fundamentally different from the GRACE-based assimilation setting. GRACE-type studies work with coarse storage-like signals that are already filtered, aggregated, and externally constrained, whereas InSAR provides high-resolution deformation observations that are noisier, more heterogeneous, and affected by mixed geophysical processes. Consequently, inversion strategies developed for coarse storage time series are not directly applicable to InSAR deformation [7], [10], [16]. Moreover, the observations and model states are not of the same physical kind: InSAR measures deformation, while W3RA provides storage states. For this reason, instead of regressing storage against storage, we formulate a deformation-space Bayesian prior in which W3RA compartments are converted into deformation predictors through elastic loading and poroelastic response operators.

This shift to using InSAR as the primary observational

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signal is scientifically important because multi-temporal InSAR resolves millimeter-scale land deformation at high spatial resolution, making it attractive for studying local hydrologic loading, aquifer compaction, and groundwater-related deformation signals that are inaccessible to GRACE and only sparsely constrained by GNSS [8], [16]–[18]. Yet translating deformation into groundwater storage remains a difficult inverse problem: it is spatially heterogeneous, temporally complex, often nonlinear, and strongly affected by subsurface structure and mixed signal sources [1], [16], [19]. Existing studies address parts of this problem but not the full workflow required here. Classical geodetic approaches use elastic and poroelastic Green’s functions to relate deformation to hydrologic mass change [1], [16], [20], [21]. Machine-learning studies have linked InSAR deformation to groundwater level, withdrawal, or aquifer parameters [22]–[24]. At the same time, physically constrained deep learning and reproducible modelling workflows are becoming more important across environmental modelling [25]–[28]. What remains missing is a compact framework that first tests identifiability under known hydro-mechanical physics, then transfers a Bayesian prior into deformation space, and only then uses learning to recover structured residuals that the Bayesian layer alone cannot separate.

The present study addresses that gap by combining a deformation-space Bayesian state-space inversion with physics-aware transformer-based residual learning in a staged framework. Stage 1 transfers the dynamic Bayesian model into deformation space and provides a stable physics-based posterior prior. Synthetic experiments show that this Bayesian stage can recover a strong posterior state under controlled clean and noisy settings. Stage 2 is introduced to learn the structured residuals that remain unresolved by Stage 1 using a Swin-based spatiotemporal residual model [29]. Stage 3 enforces deformation consistency by mapping the corrected state back through the same forward hydro-mechanical operator. In the applied Bologna case, the strongest supported real-data result is a grouped multisensor Bayesian posterior rather than a fully resolved layered inversion, and this grouped posterior is independently supported by external groundwater-well observations. Together, these results show that high-resolution layered hydrologic storage inference from InSAR deformation is plausible within a physics-aware hybrid framework, while also defining clearly where full layered real-data recovery remains challenging.

The framework therefore establishes a physically grounded and externally supported path toward layered hydrologic inversion from InSAR deformation, even though fully resolved real-data layer separation remains a longer-term objective. The paper is organized as follows. Section II introduces the study area and datasets, Section III presents the general and applied formulations, Section IV describes the synthetic and real-data workflows, Section V reports the main results, and Section VI discusses their implications, and Section VII concludes the paper.

II. STUDY AREA AND DATASETS

The applied study domain is the Bologna overlap area extracted from the broader Emilia–Romagna region in northern Italy. In the corrected MintPy/W3RA overlap used for the final grouped inversion, the working bounding box is longitude 10.2 to 12.5 and latitude 43.8 to 45.9. The shared overlap is represented on the native W3RA grid of 22×24 cells over the period 2017-01-04 to 2024-08-01, yielding 362 shared dates [5], [6]. Figure 1 is reserved for the study-area map, overlap grid, and validation-well locations. For the reader, the essential data products and their role in the workflow are summarized in Table I.

The hydrogeologic interpretation is not based on a single homogeneous groundwater body. Instead, validation uses two complementary categorizations: depth classes (shallow, intermediate, deep) and hydrogeologic groups from the ARPAE monitoring framework, including *Freatico di pianura fluviale*, *Pianura Alluvionale Appenninica – confinato superiore*, *Conoide Reno-Lavino – libero*, and related confined and alluvial-cone units. These groups become important later because the grouped posterior is judged not only by fit statistics, but also by whether it behaves consistently across hydrogeologically meaningful well subsets.

The main InSAR observation stream is a Bologna MintPy time-series product derived from Sentinel-1 and stored locally as `timeseries_SET_ERA5_ramp_demErr.h5`. The corrected product used here contains 394 acquisitions and 555 interferograms in descending geometry and includes solid-Earth-tide correction, ERA5 correction, DEM-error correction, and quadratic deramping. The product was aggregated from its native scene to the W3RA grid and converted to anomalies before assimilation. The W3RA source product is the local NetCDF model run `W3RA_2010_2024.nc`, from which the five layers *S0*, *Ss*, *Sd*, *Sg*, *Sr* were extracted and aligned to the shared period. The overlap construction is cited in the workflow described in Section IV; see also Figure 2. Table I reports the product names, coverage, variables, and preprocessing for all data streams.

GRACE/GRACE-FO is used here as a coarse regional external storage constraint through the JPL RL06.3 CRI mascon product [30], [31]. In the Bologna workflow, `lwe_thickness` anomalies were extracted as regional means over the overlap bounding box and aligned to the 362-date overlap timeline. SMAP is used as a near-surface hydrologic constraint via the NSIDC enhanced L3 radiometer daily soil-moisture product `SPL3SMP_E` [32]. One daily granule was queried per InSAR date, regional means were computed over the overlap box, and 254 valid dates survived the final alignment. SWOT is included as an exploratory external surface-water constraint using the RiverSP reach and LakeSP products [33], [34]. Over the Bologna overlap subset, 33 river times and 34 lake times were prepared, but only 9 matched dates per stream survived the final nearest-date fusion, which is why SWOT remains exploratory in the present paper rather than a main driver of the best result. Finally, independent groundwater-well observations from the ARPAE manual and automatic networks are used for validation. After

TABLE I
SUMMARY OF THE MAIN DATASETS USED IN THIS STUDY. TABLE I IS CITED IN SECTIONS II AND IV WHEN THE WORKFLOW AND PREPROCESSING ARE INTRODUCED.

Dataset	Product / file used in this study	Source	Coverage used here	Variables used	Role in study	Processing before a tion
InSAR	timeseries_SET_ERA5_ramp	Local MintPy / Sentinel-1	2017-01-04 to 2025-06-27; 394 acquisitions	deformation time series	Main observation stream	MintPy inversion, a tion to W3RA grid, a conversion
InSAR stack	inputs/ifgramStack.h5	Local MintPy inputs	555 interferograms	unwrapped phase, coherence	Documents network construction	Used to report network erties
W3RA	W3RA_2010_2024_.nc and overlap products	Local W3RA model run	overlap with InSAR: 2017-01-04 to 2024-08-01	S_0, S_s, S_d, S_g, S_r	Prior hydrologic state	overlap extraction, ding, anomaly genera
GRACE	JPL RL06.3 CRI mascon	NASA / JPL	aligned to 362 overlap dates	lwe_thickness anomalies	Regional storage constraint	Regional mean over l bounding box, a alignment
SMAP	SPL3SMP_E matched granules	NASA / NSIDC	358 matched downloads, 254 valid dates	surface soil moisture	External shallow-hydrologic constraint	One daily granule SAR date, regional over bounding box
SWOT	RiverSP reach 2.0 and LakeSP obs 2.0	NASA / PO.DAAC	River 33 times; lake 34 times; 9 matched dates per stream after fusion	river/lake surface-water summaries	Exploratory surface-water constraint	bbox subset and near matching with 14-da ance
Wells	ARPAE groundwater level tables	ARPAE Emilia-Romagna	2009-01-01 to 2024-12-18; validation overlap 2017-01-04 to 2024-08-01	piezometric head, depth to water	Independent validation	metadata merge, U lon/lat conversion, an space matching

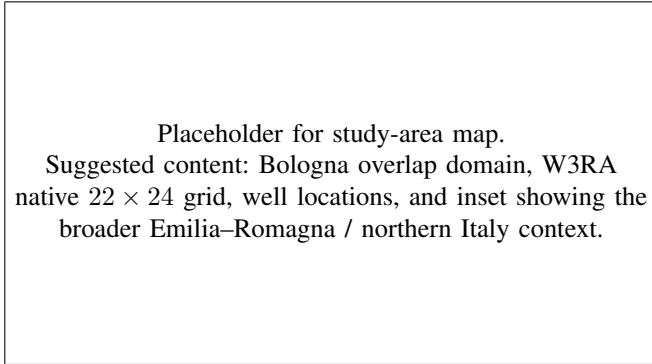


Fig. 1. Study area and validation setting. Figure 1 should show the Bologna overlap domain used in the grouped inversion, the native shared W3RA grid, and the ARPAE well network used for external validation. The working bounding box is longitude 10.2–12.5 and latitude 43.8–45.9, and the shared MintPy/W3RA overlap spans 2017-01-04 to 2024-08-01.

metadata merge, UTM-to-lon/lat conversion, anomaly construction, and temporal filtering, 83 well time series were retained for overlap-period validation. Their role is central in the interpretation of the grouped groundwater posterior, as discussed in Section V.

III. METHOD

A. General five-layer formulation

Let

$$\mathbf{x}_{t,p} = \begin{bmatrix} S_{0,t,p} \\ S_{s,t,p} \\ S_{d,t,p} \\ S_{g,t,p} \\ S_{r,t,p} \end{bmatrix} \in \mathbb{R}^5 \quad (1)$$

denote the W3RA storage state at time t and pixel p . Because InSAR measures deformation rather than storage, the W3RA

state is first mapped through an elastic–poroelastic forward model. The load-driven contribution is

$$\Delta \ell_t = \rho_w (S_0 + S_s + S_d + S_r)_t, \quad (2)$$

while the groundwater-pressure contribution is

$$\Delta p_t = \rho_w g \frac{S_{gt}}{S_{\text{eff}}}. \quad (3)$$

The corresponding displacement branches are

$$\begin{aligned} u_t^{\text{load}} &= G_{\text{load}} * \Delta \ell_t, \\ u_t^{\text{poro}} &= G_{\text{poro}} * \Delta p_t, \end{aligned} \quad (4)$$

so that the total modeled deformation is

$$u_t^{\text{tot}} = u_t^{\text{load}} + u_t^{\text{poro}}. \quad (5)$$

For the Bayesian formulation, we construct a component-wise deformation design vector

$$\mathbf{Z}_{t,p} = \begin{bmatrix} u_{t,p}^{(S_0)} & u_{t,p}^{(S_s)} & u_{t,p}^{(S_d)} & u_{t,p}^{(S_g)} & u_{t,p}^{(S_r)} \end{bmatrix}, \quad (6)$$

where the four load-driven compartments are propagated through G_{load} and the groundwater compartment through G_{poro} . Stage 1 then uses the deformation-space observation model

$$\mathbf{Y}_{t,p}^{\text{InSAR}} = \mathbf{Z}_{t,p} \boldsymbol{\theta}_{t,p} + \varepsilon_{t,p}, \quad \varepsilon_{t,p} \sim \mathcal{N}(0, R_{t,p}), \quad (7)$$

with temporal evolution

$$\boldsymbol{\theta}_{t,p} = \boldsymbol{\theta}_{t-1,p} + \boldsymbol{\eta}_{t,p}, \quad \boldsymbol{\eta}_{t,p} \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_{t,p}). \quad (8)$$

The Stage 1 posterior prior state is

$$\mathbf{x}_{t,p}^{\text{prior}} = \boldsymbol{\theta}_{t,p} \odot \mathbf{z}_{t,p}, \quad (9)$$

where $\mathbf{z}_{t,p}$ is the original W3RA state and $\odot$ denotes element-wise multiplication.

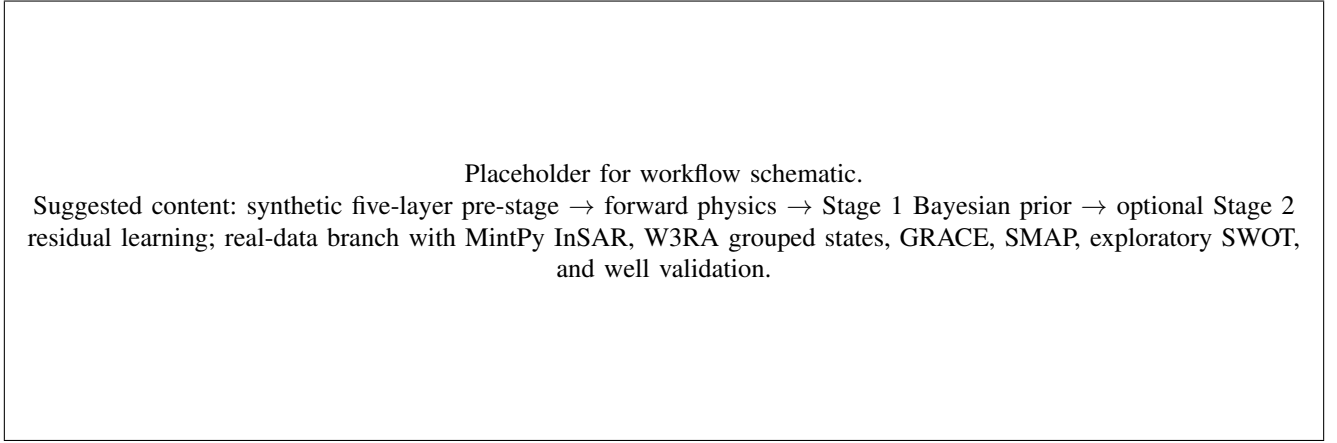


Fig. 2. Overall workflow used in this study. Figure 2 should distinguish the synthetic five-layer validation block from the applied Bologna estimator. The synthetic branch validates the deformation-space inversion machinery under known truth, whereas the real-data branch uses the grouped state in (14), assimilates InSAR, GRACE, and refreshed SMAP (with SWOT as an exploratory add-on), and validates the grouped groundwater posterior against independent ARPAE well anomalies.

Stage 2 does not replace Stage 1. It learns the residual

$$\mathbf{r}_{t,p} = \mathbf{x}_{t,p} - \mathbf{x}_{t,p}^{\text{prior}} \quad (10)$$

from local spatiotemporal context,

$$\hat{\mathbf{r}}_{t,p} = f_{\phi} \left(\mathbf{I}_{t-w+1:t, \mathcal{N}(p)}, \mathbf{C}_{\mathcal{N}(p)}, \mathbf{x}_{t, \mathcal{N}(p)}^{\text{prior}}, \Sigma_{t, \mathcal{N}(p)}^{\text{prior}} \right), \quad (11)$$

and the corrected state is

$$\mathbf{x}_{t,p}^{\text{final}} = \mathbf{x}_{t,p}^{\text{prior}} + \hat{\mathbf{r}}_{t,p}. \quad (12)$$

The final estimate must remain deformation-consistent through

$$\hat{Y}_{t,p}^{\text{InSAR}} = \mathcal{F}(\mathbf{x}_{t,p}^{\text{final}}). \quad (13)$$

B. Applied grouped Bologna estimator

The full five-layer formulation above is the general framework and the synthetic validation setting. However, the real Bologna problem is not practically well-conditioned in unconstrained five-layer form. The validated applied estimator is therefore a three-component grouped balanced multisensor state-space model with state

$$\mathbf{x}_t^{\text{grp}} = \begin{bmatrix} \text{ShallowLoad}_t \\ \text{DeepLoad}_t \\ \text{Groundwater}_t \end{bmatrix} = \begin{bmatrix} S0_t + Ss_t \\ Sd_t + Sr_t \\ Sg_t \end{bmatrix}. \quad (14)$$

This grouping preserves a dedicated groundwater component while improving numerical conditioning and physical interpretability. In the current implementation, the grouped W3RA prior is corrected through a balanced Kalman-type state-space model driven by four possible observation streams: InSAR deformation anomalies, GRACE regional anomalies, refreshed SMAP surface soil moisture, and exploratory SWOT summaries. Figure 2 should summarize both the full five-layer framework and the applied grouped Bologna implementation.

IV. EXPERIMENTS AND PROCESSING WORKFLOW

The recommended paper logic is to first present the study area and datasets (Section II), then explain how the overlap products were built, then summarize the synthetic validation block, and only after that present the grouped multisensor real-data inversion and the independent well validation. This ordering makes the workflow transparent and is the same ordering followed in Figure 2 and Table I.

A. Overlap construction and preprocessing

The Bologna overlap construction begins from the MintPy time-series product and the local W3RA model run. The corrected MintPy scene used in this study contains 394 acquisitions and 555 interferograms. The deformation product was aggregated from its original scene to the native shared W3RA grid of 22×24 cells, and the shared overlap was then restricted to the period 2017-01-04 to 2024-08-01, producing 362 shared dates. W3RA preprocessing extracted the five layers $S0, Ss, Sd, Sg, Sr$, intersected them with the corrected InSAR dates, and saved both raw overlap states and anomaly overlap states. GRACE regional means, refreshed SMAP regional means, and exploratory SWOT summaries were aligned to the same overlap timeline, although only 9 matched SWOT dates per stream survived the final bundle fusion. The complete product-level summary is already given in Table I.

B. Synthetic validation workflow

The synthetic stage is a pre-stage validation block, not the final applied result. It verifies that the inversion machinery can recover known signals under model-consistent conditions. Smooth W3RA-like storage fields are generated for $S0, Ss, Sd, Sg, Sr$, converted to deformation-space predictors using the same forward physics later used in the real inversion, and then used to construct synthetic observations of the form

$$Y_{t,p} = Z_{t,p} \theta_{t,p} + \varepsilon_{t,p}. \quad (15)$$

Stage 1 MCMC is first tested for posterior recovery, after which Stage 2 is tested only as a residual-learning refinement

TABLE II

SYNTHETIC AND REAL EXPERIMENT BLOCKS USED IN THIS STUDY. TABLE II IS CITED IN SECTION IV TO CLARIFY THE DISTINCTION BETWEEN THE GENERAL FRAMEWORK, THE APPLIED GROUPED ESTIMATOR, AND THE VALIDATION STAGES.

Block	State formulation	Observation streams	Output
Synthetic Stage 1	five-layer (S_0, S_s, S_d, S_g, S_r)	synthetic deformation	posterior prior recovery
Synthetic Stage 2	five-layer residual refinement	synthetic deformation + prior	residual-learning diagnostic
Real grouped Stage 1	grouped (S_0+S_s, S_d+S_r, S_g)	InSAR + GRACE SMAP + exploratory SWOT	validated applied posterior
External validation	grouped groundwater vs wells	ARPAE well anomalies	lag-aware anomaly correlation

Placeholder for synthetic validation figure.

Suggested content: truth vs posterior summaries for Stage 1, deformation reconstruction skill, and a compact note that Stage 2 is technically feasible but not the main driver of the real-data result.

Fig. 3. Synthetic validation summary. Figure 3 should show that the five-layer deformation-space inversion machinery works strongly under model-consistent synthetic conditions and serves as a pre-stage validation block before the grouped real-data application.

under known truth. The synthetic setup uses 24 time steps, a 12×12 grid, five layers, and a clean reference run with $\text{noise_scale} = 0.0$. The key synthetic result is that the inversion machinery works strongly in a model-consistent synthetic setting, but synthetic success alone does not guarantee identifiability in real data. Figure 3 is reserved for a compact synthetic summary and is cited here because the synthetic stage is part of the methodological justification of the paper.

C. Grouped multisensor Bologna inversion

For the real Bologna application, the final workflow is: process the MintPy time series, aggregate the product to the native W3RA grid, build the shared overlap period, prepare external constraints (GRACE, SMAP, exploratory SWOT, and wells), run the grouped balanced multisensor Stage 1 model in (14), and validate the grouped groundwater posterior against wells. The grouped real-data model uses a balanced Kalman-type state-space formulation in which the grouped W3RA prior is corrected through multiplicative state factors. SWOT was successfully prepared and tested, but it does not materially improve the current Bologna grouped inversion because the matched time overlap with the other streams is too sparse. This point should be visible later in the Results section and should not be overstated in the paper.

V. RESULTS

The strongest real-data result obtained in this study is not the original unconstrained layered inversion, but a grouped balanced multisensor Stage 1 model over the corrected Bologna MintPy/W3RA overlap period from 2017-01-04 to 2024-08-01. The grouped posterior state is given by (14) and is inferred by combining InSAR deformation anomalies, GRACE regional anomalies, and refreshed SMAP surface soil moisture in a balanced state-space formulation. Figure 4 is reserved for the main grouped posterior maps and regional fit summary, and Table III summarizes the central quantitative metrics.

On the corrected overlap run, the grouped posterior improves all three observation streams relative to the grouped prior. Regional posterior skill reaches approximately $R^2 \approx 0.971$ for InSAR, $R^2 \approx 0.487$ for GRACE, and $R^2 \approx 0.618$ for SMAP, while at the tiled level the grouped posterior achieves an InSAR posterior $R^2 \approx 0.905$. An important practical outcome is that posterior grouped magnitudes remain physically plausible rather than exploding numerically: *ShallowLoad* has an absolute maximum of about 52.1 mm, *DeepLoad* about 88.1 mm, and *Groundwater* about 48.1 mm. This is a major improvement over the earlier unconstrained real-data layered inversions, which were numerically unstable and physically uninterpretable.

The grouped posterior was then tested against independent Bologna groundwater wells. Since the model output is a storage-like grouped posterior and the wells measure hydraulic head, the comparison was performed in anomaly space rather than by direct one-to-one unit equality. The validation used spatial interpolation from the grouped Kalman tile grid to each well, nearest-date temporal alignment, and a lag search over $-90, -60, -30, -14, 0, 14, 30, 60, 90$ days. Under this setup, 83 well time series were evaluated, all 83 had positive best-match anomaly correlation, 74 had correlation at least 0.3, 62 had correlation at least 0.5, the median correlation was about 0.704, and the top-10 mean was about 0.947. Depth-stratified validation gives a physically interpretable pattern, with median correlations of about 0.767 for shallow wells, 0.731 for intermediate wells, and 0.617 for deep wells. Figure 5 should summarize these well-validation results and the trusted validation panel.

When wells are allowed to match the best-performing grouped state after lag search, *Groundwater* is the best state for 42 stations, *ShallowLoad* for 25, and *DeepLoad* for 16. To avoid over-interpreting weaker hydrogeologic settings, a conservative trusted validation subset was defined, containing 6 hydrogeologic groups and 21 stations. This trusted subset is the strongest external evidence supporting the grouped groundwater posterior and should therefore anchor the interpretation of the applied result.

VI. DISCUSSION

Detailed mathematical proofs, forward-operator derivations, conditioning diagnostics, and extended validation results are provided in the Supplementary Material. The paper supports three main conclusions. First, the synthetic five-layer pre-stage shows that the deformation-space inversion machinery works

Placeholder for main grouped-result figure.
Suggested content: grouped posterior maps for ShallowLoad, DeepLoad, and Groundwater; regional observed-vs-prior-vs-posterior timeseries for InSAR, GRACE, and SMAP; note that SWOT is exploratory and not a major driver of the best result.

Fig. 4. Main grouped Bologna result. Figure 4 should show the grouped posterior maps and the internal multisensor fit summary for InSAR, GRACE, and SMAP over the shared overlap period. This is the central applied-result figure of the paper.

TABLE III
MAIN QUANTITATIVE RESULTS FOR THE GROUPED BOLOGNA ESTIMATOR.
TABLE III IS CITED IN SECTION V WHEN THE INTERNAL MULTISENSOR FIT AND EXTERNAL WELL VALIDATION ARE SUMMARIZED.

Metric block	Value
Regional InSAR posterior R^2	≈ 0.971
Regional GRACE posterior R^2	≈ 0.487
Regional SMAP posterior R^2	≈ 0.618
Tiled InSAR posterior R^2	≈ 0.905
Max ShallowLoad	52.1 mm
Max DeepLoad	88.1 mm
Max Groundwater	48.1 mm
Evaluated wells	83
Median well correlation	0.704
Wells with corr. ≥ 0.3	74
Wells with corr. ≥ 0.5	62
Trusted hydrogeologic groups	6
Trusted stations	21

under model-consistent controlled conditions, so the core Bayesian formulation is justified as an inversion engine rather than only as a heuristic fit. Second, the unconstrained real-data five-layer inversion is not practically stable in the Bologna case, which is why the grouped balanced multisensor Stage 1 formulation in (14) becomes the validated applied estimator. Third, the grouped posterior is not merely an internal InSAR-fitting artifact: it is physically plausible in magnitude, fits InSAR, GRACE, and SMAP jointly, and is independently supported by well-head anomalies across the Bologna network.

These conclusions also clarify the role of SWOT and Stage 2 in the current paper. SWOT is a real data stream in the workflow and should appear in Table I and in the methodological description, but it should not be overstated in the results because only 9 matched dates per stream survive the current overlap fusion. Similarly, Stage 2 remains theoretically

Placeholder for well-validation figure.
Suggested content: summary panel of best-match correlations, trusted-wells map, and compact depth-class summary.

Fig. 5. Independent well validation of the grouped groundwater posterior. Figure 5 should summarize the lag-aware anomaly correlation against ARPAE wells and highlight the conservative trusted subset of stations and hydrogeologic groups used for interpretation.

justified and useful as a nonlinear residual-learning module, but it is not yet the main applied estimator in the Bologna case. The present real-data evidence supports it as an exploratory lag/mismatch learner rather than an operational improvement over the grouped Stage 1 posterior.

VII. CONCLUSION

The paper presents a hybrid hydrologic-inversion framework in which a full five-layer deformation-space Bayesian formulation is retained as the general model and synthetic validation setting, while the Bologna real-data application is solved with a grouped balanced multisensor Stage 1 estimator. The applied study domain covers longitude 10.2–12.5 and latitude 43.8–45.9 on a native 22×24 overlap grid over the period 2017-01-04 to 2024-08-01. The current best real-data result is the grouped posterior $[S_0 + S_s, S_d + S_r, S_g]$, obtained from InSAR deformation anomalies, GRACE regional anomalies, and refreshed SMAP surface soil moisture, with exploratory SWOT integration and independent validation against ARPAE wells. The grouped posterior remains physically sane in scale and is supported by a strong well-anomaly validation panel, including a conservative trusted subset of 21 stations across 6 hydrogeologic groups. The full layered real-data inversion

is therefore not the validated result yet; it remains the longer-term objective, with grouped multisensor Bayesian inference providing the currently supported operational estimator.

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